

A four-point counterexample to the verbose pullback triangle inequality

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Abstract

The fixed-degree pullback matching distance of verbose Vietoris–Rips barcodes fails the triangle inequality. In degree one, three four-point ultrametric spaces, with nonzero distances only 1 and 2, have pairwise distances 0, 0, 1. More generally, in every degree $k \geq 1$ and over every field, the same phenomenon occurs on three spaces of equal cardinality $k + 3$. Every barcode involved is nonempty. A rank-counting argument proves the positive lower bound over *all* finite common pullbacks, not merely an enumerated set. The common cardinality $k + 3$ is minimal. The degree-one zero-distance pair used here is already in Mémoli–Zhou, Example 5.10; the third space and the uniform lower bound complete a counterexample.

Evidence status. Complete proposed disproof, awaiting independent mathematical review. The supplied proof and programs are AI-generated. Codex and separate AI reviewers checked the argument, programs, and target sources during submission preparation; details appear in `verification_record.md`. No human peer review, formal proof-assistant verification, or repository acceptance is claimed.

1 The target and conventions

The target is original AIM 544, *Triangle inequality for the pullback distance of verbose persistence barcodes*, pinned at `aa776a0` [1]. As of 2 October 2026 it is AIM 534 at `fa98b75`; only the numbered heading changed, and the target remains Open. Full revision hashes and the dated prior-submission audit are in `target_audit.md`. No overlapping claim was found in the inspected public records.

Fix a field $\mathbb{F}$. For a finite nonempty pseudometric space (S, d) , give each simplex its diameter as filtration value. Write $V_k(S, d)$ for the degree- k barcode obtained by the ordinary boundary reduction *retaining all zero-length pairs*. Endpoint pairs have their full multiplicities. For equal-size finite barcodes, m is the least maximum ℓ^∞ endpoint discrepancy over bijections; no additional diagonal points are allowed. For finite metric spaces A, B , the target distance is

$$D_k(A, B) = \inf_{S \text{ finite}; p: S \rightarrow A; q: S \rightarrow B} m(V_k(S, p^*d_A), V_k(S, q^*d_B)). \quad (1)$$

Distinct vertices at zero pullback distance are not identified. We work only in degrees $k \geq 1$, so all bars have finite endpoints. On a common set of s vertices, either pullback has exactly $\binom{s-1}{k+1}$ degree- k pairs, since its final complex is the full simplex. Thus the compared barcodes have equal cardinality (Lemma 2). The published fixed-degree question is in Remark 1.1 and Definition 5.3 of [2].

2 The four-point counterexample

Take three copies of $\{1, 2, 3, 4\}$ with distance matrices

$$d_X = \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 0 \end{pmatrix}, \quad d_Y = \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 \\ 2 & 2 & 1 & 0 \end{pmatrix}, \quad d_Z = \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 0 \end{pmatrix}. \quad (2)$$

All are genuine ultrametrics. At scale 1, the clusters of X have sizes 3, 1, those of Y have sizes 2, 2, and those of Z have sizes 1, 1, 1, 1. At scale 2, all four vertices form a simplex.

Theorem 1. *For the spaces in (2), over every field,*

$$\boxed{D_1(X, Y) = 0, \quad D_1(Y, Z) = 0, \quad D_1(X, Z) = 1.}$$

In particular, these spaces disprove the universal assertion in AIM 544.

We first give the elementary counting facts that justify every barcode and the lower bound over arbitrary pullbacks.

Lemma 2 (Boundary ranks of a simplex). *For the full simplex on $r \geq 1$ vertices and $k \geq 1$,*

$$\text{rank } \partial_{k+1} = \binom{r-1}{k+1},$$

where a binomial coefficient is zero when its lower index exceeds its nonnegative upper index. For a disjoint union of simplexes on $r_1, \dots, r_s$ vertices, the rank is the sum of these quantities.

Proof. Fix one vertex v . The boundaries of the $(k+1)$ -simplexes containing v are independent: each has a distinct face of dimension k not containing v , with coefficient 1 or -1 . They span the boundary space. Indeed, the boundary of a $(k+1)$ -simplex not containing v is a linear combination of these boundaries, obtained by applying $\partial^2 = 0$ to its cone with vertex v . There are $\binom{r-1}{k+1}$ such simplexes. The argument works in every characteristic. Disjoint components give a direct sum. $\square$

Lemma 3 (Counting the diagonal bars). *Suppose a finite simplicial filtration changes only at $t_0 < \dots < t_s$, and its positive-degree homology vanishes at every threshold. Then every positive-degree verbose bar is diagonal. The number of degree- k bars with death time at most t_i is $\text{rank } \partial_{k+1}(K_{t_i})$. Consequently, the multiplicity of (t_i, t_i) is the increment of this rank from the preceding threshold.*

Proof. A positive-length bar would give nonzero homology at some threshold or on an interval between successive thresholds. Thus no such bars occur. In a boundary reduction, nonzero columns of degree $k+1$ are exactly the deaths of degree- k pairs. Reducing all columns with entry time at most t_i gives as many nonzero columns as the rank of the corresponding boundary map. This statement includes deaths paired with births at the same filtration value. Differences count deaths at the individual thresholds. $\square$

2.1 The two zero distances

The filtrations here are disjoint unions of simplexes at every scale, so Lemmas 2–3 give

$$V_1(X) = \{(1, 1), (2, 2), (2, 2)\}, \quad V_1(Y) = V_1(Z) = \{(2, 2)\}^3. \quad (3)$$

Use the original four-element set, with bijections onto Y and Z . Their equal barcodes prove $D_1(Y, Z) = 0$.

For X, Y , use a five-element set $S = \{1, 2, 3, 4, 5\}$ and the surjections

$$\begin{array}{c|ccccc} s & 1 & 2 & 3 & 4 & 5 \\ \hline p(s) \in X & 1 & 2 & 3 & 4 & 4 \\ q(s) \in Y & 1 & 2 & 3 & 4 & 1 \end{array} \quad (4)$$

The first pullback repeats the isolated point of X ; the second repeats one member of a close pair in Y . Their cluster sizes at the thresholds are

$$\begin{array}{c|ccc} & 0 & 1 & 2 \\ \hline (S, p^*d_X) & 2, 1, 1, 1 & 3, 2 & 5 \\ (S, q^*d_Y) & 2, 1, 1, 1 & 3, 2 & 5 \end{array}$$

For degree one, the boundary ranks at these thresholds are 0, 1, 6. Hence both pullbacks have barcode

$$\{(1, 1)\} \sqcup \{(2, 2)\}^5. \quad (5)$$

This is a zero-cost admissible matching, proving $D_1(X, Y) = 0$.

2.2 A lower bound for every common pullback

Let S be *any* finite set surjecting onto X and Z . Let $a, b, c, d \geq 1$ be the fiber sizes for its map to X . At threshold 0, the X -pullback is the disjoint union of simplexes on a, b, c, d vertices. At threshold 1, these merge into simplexes on $a + b + c$ and d vertices. The number of $(1, 1)$ bars is therefore

$$\begin{aligned} & \binom{a+b+c-1}{2} + \binom{d-1}{2} - \binom{a-1}{2} - \binom{b-1}{2} - \binom{c-1}{2} - \binom{d-1}{2} \\ &= ab + ac + bc - 2 \geq 1. \end{aligned} \quad (6)$$

This includes all possible duplication patterns, without any size bound.

The Z -pullback has only distances 0 and 2. At threshold 0 it is a disjoint union of simplexes, and at threshold 2 it is a full simplex. Its degree-one bars are therefore only $(0, 0)$ and $(2, 2)$. Every bijection must match the $(1, 1)$ bar supplied by (6) to one of these two pairs. Either choice has cost 1. Thus every admissible comparison in (1) costs at least 1.

Conversely, the original four-point barcodes in (3) can be matched with cost 1. This proves $D_1(X, Z) = 1$ and completes the proof of Theorem 1. $\square$

Attribution and scope. The spaces X, Y , their degree-one barcodes, and a zero-cost duplication comparison already occur in Mémoli–Zhou, Example 5.10 and Figure 7, after relabeling [2]. No novelty is claimed for that pair. The choice of an equilateral third space with distance 2, together with the uniform obstruction (6), completes the proposed disproof. Our Z differs from the equilateral distance-1 space denoted Z in that source example. This is a fixed-degree argument, not the source’s distinct all-degree pullback-interleaving counterexample. It does not depend on any convention for empty barcodes: every original barcode here has three entries and every common pullback has at least four vertices.

3 A counterexample in every positive degree

Theorem 4. *For every field $\mathbb{F}$ and every integer $k \geq 1$, there are three ultrametric spaces X_k, Y_k, Z_k , each with $k + 3$ points and nonzero distances in $\{1, 2\}$, such that*

$$D_k(X_k, Y_k) = D_k(Y_k, Z_k) = 0, \quad D_k(X_k, Z_k) = 1.$$

The degree- k barcode of every original space and every common pullback in these comparisons is nonempty.

Proof. Set $n = k + 3$. Give X_k a close cluster of size $k + 2$ and a singleton, and give Y_k two close clusters of sizes $k + 1$ and 2. Distances between distinct points in the same close cluster are 1; all remaining off-diagonal distances are 2. Let Z_k be the equilateral n -point space with distance 2.

Lemmas 2–3 give

$$V_k(X_k) = \{(1, 1)\} \sqcup \{(2, 2)\}^{k+1}, \quad (7)$$

$$V_k(Y_k) = V_k(Z_k) = \{(2, 2)\}^{k+2}. \quad (8)$$

In particular, the original sets give a zero-cost comparison of Y_k, Z_k .

Repeat the singleton in X_k and one vertex of the size- $(k + 1)$ cluster in Y_k . Both pullbacks have $k + 4$ vertices. At 0, one fiber has size 2 and all others have size 1; since $k \geq 1$, this yields no degree- k bars at 0. At 1, both have component sizes $k + 2, 2$, with boundary rank 1. At 2, the rank is $\binom{k+3}{k+1}$. Both barcodes are

$$\{(1, 1)\} \sqcup \{(2, 2)\}^{\binom{k+3}{k+1}-1}, \quad (9)$$

so $D_k(X_k, Y_k) = 0$.

Now consider any pullback of X_k . Let $m_1, \dots, m_{k+2} \geq 1$ be the fiber sizes over its large cluster, and put $M = \sum_i m_i$. The singleton fiber's contribution cancels between levels 0 and 1. The multiplicity of $(1, 1)$ is

$$B = \binom{M-1}{k+1} - \sum_{i=1}^{k+2} \binom{m_i-1}{k+1} \geq 1. \quad (10)$$

For a direct proof of the inequality, form disjoint sets A_i with $|A_i| = m_i - 1$ and a further disjoint set R of size $k + 1$. Their union has size $M - 1$. The first binomial counts all its $(k + 1)$ -element subsets; the sum subtracts just those lying entirely in some A_i . The subset R is not subtracted, so the remainder is at least one.

Every pullback of Z_k has only $(0, 0)$ and $(2, 2)$ bars, as in the degree-one argument. Equation (10) forces matching cost at least 1 for every common pullback. The original barcodes in (7) give cost 1, proving equality. Finally, on any $s \geq k + 3$ vertices a full simplex has $\binom{s-1}{k+1} > 0$ degree- k pairs, which proves nonemptiness. $\square$

4 Minimal equal cardinality and consequences

Lemma 5 (A diameter bar). *Let A be a finite pseudometric space with $r \geq k + 2$ distinct vertices and diameter δ , where $k \geq 1$. Then $V_k(A)$ contains (δ, δ) .*

Proof. Choose two vertices realizing the diameter and extend them to a set of $k + 2$ vertices, defining a $(k + 1)$ -simplex σ . The cycle $\partial\sigma$ has a nonzero coefficient on a k -face containing those two vertices: delete any of the other k vertices. That face has entry time δ . Thus $\partial\sigma$ is not in the chain space strictly below δ , and the dimension of the space of k -cycles increases at δ .

The number of degree- k barcode births through a threshold is the number of zero reduced degree- k columns, namely the dimension of that threshold's k -cycle space. Hence at least one bar is born at δ . The filtration is the full simplex at δ , so every bar born then also dies then. If $\delta = 0$, the same argument uses the empty complex below 0. $\square$

Proposition 6. *Among triples whose three spaces have the same cardinality, $k + 3$ is the smallest cardinality at which the triangle inequality for D_k can fail.*

Proof. For equal cardinality $n \leq k + 1$, the original degree- k barcodes are empty, so D_k is identically zero on that cardinality class, using the repository's explicit empty-matching convention.

For $n = k + 2$, the original degree- k barcode has exactly one pair by Lemma 2. Lemma 5 identifies it as $(\text{diam } A, \text{diam } A)$. This gives

$$D_k(A, B) \leq |\text{diam } A - \text{diam } B|.$$

Every admissible common pullback has at least $k + 2$ vertices, has the same diameter as its target, and has a diameter bar by Lemma 5. All endpoints in the other barcode are at most its target's diameter. Matching the larger diameter bar therefore costs at least $|\text{diam } A - \text{diam } B|$, proving equality. The absolute-value triangle inequality applies on this cardinality class. Theorem 4 attains failure at $n = k + 3$. $\square$

Consequences. In every positive degree, the zero-distance relation is not transitive. Moreover, there is no finite constant C such that

$$D_k(A, C') \leq C(D_k(A, B) + D_k(B, C'))$$

for all finite metric spaces: the right-hand side for our triple is zero. Multiplying all distances by $a > 0$ gives distances $0, 0, a$, so there is no universal additive bound on the triangle defect either. Taking an infimum of sums along chains produces a *different* quantity, which identifies X_k and Z_k at zero cost.

The witness also works if common pullbacks are restricted to *correspondences*, meaning $s \mapsto (p(s), q(s))$ is injective. In the general construction, take identity labels for the first n vertices and an additional pair $(n, 1)$; these $n + 1$ ordered pairs are distinct. The other two upper bounds use bijections. The lower bound already holds for all finite tripods, hence for this subclass as well.

These conclusions concern the specified fixed-degree verbose pullback construction. Ordinary positive-degree persistence barcodes of these ultrametric spaces are empty, so they give no counterexample to the usual bottleneck metric. The established degree-zero result is also unaffected.

5 Verification and research status

The proof is exact and self-contained; it does not depend on numerical optimization or a finite search over pullbacks. Two standard-library Python implementations accompany this note.

Implementation	Supporting checks	Matrix computations
<code>verify_boundary.py</code>	6,653	720 persistence reductions
<code>verify_ranks.py</code>	3,650	356 boundary ranks

The first program implements oriented boundary-column reduction over $\mathbb{Q}, \mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$, retaining all zero-length pairs. It checks the family for $1 \leq k \leq 6$, including two tie orders. It also tests every positive four-fiber size vector with $4 \leq |S| \leq 8$ in three fields, resulting in 5,226 pairwise comparisons. The upper bounds and endpoint obstruction are reproduced exactly. Its matching routine introduces no artificial diagonal points.

The second program does not import the first. It builds the boundary matrices at thresholds $0, 1, 2$, uses dense row elimination over $\mathbb{Q}$ and $\mathbb{F}_3$, checks vanishing positive-degree homology, and compares rank increments with the claimed barcode multiplicities. It additionally checks 3,267 finite instances of (10). These finite tests do not establish the universal quantifier; the subset-counting proof does.

Run, from the extracted package directory:

```
python3 verify_boundary.py --output boundary_results.json
python3 verify_ranks.py --output rank_results.json
```

Limitations. Separate computational implementations are not independent mathematical review. No Lean or other proof-assistant certificate is supplied. The available repository and literature audit found no matching prior 544 submission, but it is not an exhaustive certification of unpublished work or inaccessible records. The accompanying resolution report is prepared for submission by Siavash Sadeghi. Under the repository’s evidence rules [3], the appropriate status is *Solution claimed* until a documented independent audit or publication establishes a stronger status.

References

- [1] S. Brunton, M. Colbrook, M. de Hoop, G. Stepaniants, A. Townsend, and R. Ward, *AIM: Original problem 544, Triangle inequality for the pullback distance of verbose persistence barcodes*. Target at [commit aa776a0](#); [current AIM 534 statement](#) inspected 2 October 2026.
- [2] F. Mémoli and L. Zhou, *Ephemeral persistence features and the stability of filtered chain complexes*, *Journal of Computational Geometry* **15**(2), 258–328 (volume labeled 2024; published 2025). [doi:10.20382/jocg.v15i2a8](#). See Remark 1.1; Definitions 4.7, 5.3; Example 5.10 and Figure 7; and Proposition 6.2. Author revision: [arXiv:2208.11770v8](#), 21 August 2025.
- [3] *AIM contribution and evidence rules*. [CONTRIBUTING.md](#), inspected 2 October 2026.